MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

INFORMATION ON RESEARCH RESULTS

1. General information:

Project code: THS2026-68

Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events

Project coordinator: Lê Thị Ngọc Ảnh

Implementing institution: Trường Công nghệ Thông tin và Truyền thông, Đại học Cần Thơ

Duration: May–October 2026 (six months).

2. Objective(s):

Develop an AI-based mobile system to receive and analyze flood rescue information and support decision-making under unstable network conditions, with the aim of reducing loss of life and property damage.

3. Creativeness and innovativeness:

The project proposes an integrated approach combining multimodal data analysis, Edge AI and intelligent rescue-event clustering within a unified system.

Instead of relying entirely on centralized server processing, the proposed system aims to perform AI inference directly on mobile devices using optimized lightweight deep learning models, thereby reducing computational requirements and the transmission of large data files.

Furthermore, the integration of location, time and rescue information for identifying and grouping related events is intended to reduce duplicate reports and support visualization of rescue situations on a digital map.

4. Research results:

The research investigates the development of flood rescue datasets, deep learning-based image classification, urgency estimation using Logistic Regression from structured user

selections, AI model optimization for mobile devices, and the design of a rescue information management and event clustering system.

For image classification, the model was evaluated on a test set of 256 images. Using seed 1024, the model achieved 75.78% accuracy and 74.26% macro-F1, with three critical errors. The structured pruning experiment achieved 80.08% accuracy, with 4 critical errors.

On the mobile device, image classification and urgency estimation from structured form selections provide complementary information. Their outputs are sent to the server for aggregation, event clustering and priority ranking. The research also examines data synchronization and system operation under unstable network conditions.

In weak-network simulations with bandwidth of 50 kbit/s and a round-trip time of 600 ms, both metadata-only and compressed-image modes succeeded in 257/257 attempts (100%), compared with 230/257 (89.49%) for original images. The two modes transmitting less data achieved higher transmission success rates than original images under the tested conditions. Each mode was measured separately on a computer; transmission over real mobile networks requires further validation.

The available results demonstrate the feasibility of selected prototype components in a research environment. They do not yet establish improvements in real-world rescue coordination or disaster response effectiveness.

5. Products:

The project registers three main scientific and technological products:

– A mobile application supporting disaster rescue, integrating AI models and rescue request submission functions.

– A web-based rescue information management system featuring a digital map for visualizing rescue locations and clustering related events.

– A trained and optimized multimodal AI model package for analyzing rescue information and supporting deployment on mobile devices.

In addition, the project registers a research bulletin, a research summary and a demonstration video of no more than two minutes, in accordance with Can Tho University's requirements. No scientific publication is registered as an official research output.

6. Effects, technology transfer means and applicability:

The project contributes to education and research by providing practical experience in deep learning, multimodal data processing, Edge AI, mobile application development and

disaster-related information systems.

From a technological perspective, the research explores the feasibility of processing rescue information at the network edge and integrating event clustering with spatial data visualization.

The proposed system has the potential to support the collection, classification and consolidation of rescue information under limited network connectivity, thereby assisting rescue-related decision-making and resource coordination.

The project outputs are intended to be packaged and transferred to the School of Information and Communication Technology, Can Tho University, for educational, research and further development purposes.

Future practical deployment requires additional validation of multimodal analysis, mobile inference performance, network synchronization, event clustering reliability and system operation under realistic disaster conditions. Real-world rescue effectiveness and socioeconomic benefits have not yet been experimentally established.